In [3]:
```
#   Assignment: 02      Topic: decision making conditions     date: 17/08/2026
'''
1. Write a program to input electricity unit charges and calculate
total electricity bill according to the given condition:
For first 50 units Rs. 0.50/unit
For next 100 units Rs. 0.75/unit
For next 100 units Rs. 1.20/unit
For unit above 250 Rs. 1.50/unit
An additional surcharge of 20% is added to the bill
'''
unit = float(input("Enter unit: "))

if unit <= 50:
    bill = 0.50 * unit
elif unit <= 150:
    bill = 0.50 * 50 + 0.75 * (unit - 50)
elif unit <= 250:
    bill = 0.50 * 50 + 0.75 * 100 + 1.20 * (unit - 150)
else:
    bill = 0.50 * 50 + 0.75 * 100 + 1.20 * 100 + 1.50 * (unit - 250)

print("The Electricity bill is:", bill + bill * 20 / 100)
```

```
Enter unit:  300

The Electricity bill is: 354.0
```

In [4]:
```
'''
2. Write a program to check whether the last digit of a number
(entered by user) is divisible by 3 or not.
'''
n = int(input("Enter a number:"))
last_digit = n % 10
if (last_digit % 3 == 0):
    print("The number is divisible by 3")
else:
    print("The number is not divisible by 3")
```

```
Enter a number226
The number is divisible by 3
```

In [5]:
```
'''
3. Write a program to display "Hello" if a number entered
by user is a multiple of 5 and 7, otherwise print "Bye"
'''
n = int(input("Enter a number: "))
if(n % 35 == 0):
    print("Hello")
else:
    print("Bye")
```

```
Enter a number: 35
Hello
```

In [7]:
```python
'''
4. A company decided to give bonus to employee according to following criteria:
Period of Service       Bonus
More than 10 years       10%
>=6 and <=10              8%
Less than 6 years         5%
Ask user for their salary and years of service and print the net bonus amount.
'''
year = int(input("Enter the year:"))
sal = float(input("Enter the salary:"))
if (year > 10):
    sal = sal + sal * 10/100
if (year >= 6 and year <= 10):
    sal = sal + sal * 8/100
if (year < 6):
    sal = sal + sal * 5/100
print("The salary is: ", sal)
```

```
Enter the year:5
Enter the salary:50000.500
The salary is:  52500.525
```

In [9]:
```python
'''
5. Write a python program that takes a number.
If the number is divisible by 3, it should return “Fizz”.
If it is divisible by 5, it should return “Buzz”.
If it is divisible by both 3 and 5, it should return “FizzBuzz”.
Otherwise, it should return the same number.
'''
num = int(input("Enter a number: "))
if (num % 3 == 0):
    print("Fizz")
elif (num % 5 == 0):
    print("Buzz")
elif (num % 3 == 0 and num % 5 == 0):
    print("FizzBuzz")
else:
    print(num)
```

```
Enter a number: 15
Fizz
```

In [5]:
```
'''
6. Write a program for checking the speed of drivers..
    1. If speed is less than 70, it should print “Ok”.
    2. Otherwise, for every 5km above the speed limit (70),
       it should give the driver one demerit point and print the
       total number of demerit points.For example, if the speed
       is 80, it should print: “Points: 2”.
    3. If the driver gets more than 12 points, it should print:
        “License suspended”
'''
speed = int(input("Enter the speed: "))
point = (speed - 70) // 5

if (speed < 70):
    print("Ok")
else:
    if (point > 12):
        print("License suspended")
    else:
        print("Points: ", point)
```

```
Enter the speed:  135

License suspended
```

In [7]:
```
'''
7. A year is a leap year if it is divisible by 4,
except that years divisible by 100 are not leap years unless
they are also divisible by 400.Write a program that asks the
user for a year and prints out whether it is a leap year or not.
'''
year = int(input("Enter a year: "))

if (year % 400 == 0):
    print(year,"is Leap year")
elif (year % 100 == 0):
    print(year,"is Not a leap year")
elif (year % 4 == 0):
    print(year,"is Leap year")
else:
    print(year,"is Not a leap year")
```

```
Enter a year:  2024

2024 is Leap year
```

In [12]:
```
'''
8. Write a program to input marks of five subjects Physics,
Chemistry, Biology,Mathematics and Computer. Calculate percentage
and grade according to following:
    Percentage >= 90% : Grade A
    Percentage >= 80% : Grade B
    Percentage >= 70% : Grade C
    Percentage >= 60% : Grade D
    Percentage >= 40% : Grade E
'''
phy = float(input("Enter the Physics mark: "))
che = float(input("Enter the Chemistry mark: "))
math = float(input("Enter the Mathematics mark: "))
comp = float(input("Enter the Computer mark: "))
bio = float(input("Enter the Biology mark: "))
perc = ((phy + che + math + comp + bio) / 500) * 100

print("Percentage =", perc)

if perc >= 90:
    print("Grade A")
elif perc >= 80:
    print("Grade B")
elif perc >= 70:
    print("Grade C")
elif perc >= 60:
    print("Grade D")
elif perc >= 40:
    print("Grade E")
else:
    print("Grade F")
```

```
Enter the Physics mark:  87
Enter the Chemistry mark:  86
Enter the Mathematics mark:  88
Enter the Computer mark:  85
Enter the Biology mark:  78

Percentage = 84.8
Grade B
```

In [13]:
```python
'''
9.Write a program to check whether the triangle is equilateral,
isosceles or scalene triangle.
'''
a = float(input("Enter first side: "))
b = float(input("Enter second side: "))
c = float(input("Enter third side: "))

if a == b and b == c:
    print("Equilateral triangle")
elif a == b or b == c or a == c:
    print("Isosceles triangle")
else:
    print("Scalene triangle")
```

```
Enter first side:  5
Enter second side:  5
Enter third side:  5

Equilateral triangle
```

In [18]:
```python
'''
10. Write a program that calculates the ticket price based on
age with the followingconditions: age below 12 pay a ticket price
of 5, age below 18 pay a ticket price of 10,age below 60 pay a ticket
price of 20, age over 60 play a ticket price of 15.
'''
age = int(input("Enter your age: "))
if (age < 12):
    print("pay a ticket price of 5")
elif (age < 18):
    print("pay a ticket price of 10")
elif (age < 60):
    print("pay a ticket price of 20")
elif (age > 60):
    print("60 pay a ticket price of 15")
else:
    print("He is not exist")
```

```
Enter your age:  78

60 pay a ticket price of 15
```

In [20]:
```python
'''
11. An automobile company manufactures both a two wheeler (TW)
and a four wheeler (FW).A company manager wants to make the production
of both types of vehicle according to the given data below:
1st data, Total number of vehicle (two-wheeler + four-wheeler)=v
2nd data, Total number of wheels = WThe task is to find how many
two-wheelers as well as four-wheelers need to manufacture . Example :
Input : 200 -> Value of V 540 -> Value of W
Output : TW =130 FW=70
'''
v = int(input("Enter total number of vehicles: "))
w = int(input("Enter total number of wheels: "))

if (w % 2 != 0 or w < 2 * v or w > 4 * v):
    print("INVALID INPUT")
else:
    fw = (w - 2 * v) // 2
    tw = v - fw
    print("TW =", tw)
    print("FW =", fw)
```

```
Enter total number of vehicles:  200
Enter total number of wheels:  540

TW = 130
FW = 70
```